School of Electronic Engineering
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Final Report

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Project Title:
NLP model to detect fake reviews from online reviews in Telugu.

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Abstract

In the digital age, where decision-making is increasingly influenced by online content, consumers often turn to reviews to guide their choices in products and services. However, the authenticity and reliability of these online reviews are frequently compromised by the presence of biased or misleading information, especially in languages such as Telugu, where digital content moderation tools are less developed compared to English. This project introduces a novel Natural Language Processing (NLP) system, specifically designed to analyse and identify fake reviews within the Telugu-speaking online community. By utilising advanced NLP techniques, the system aims not only to distinguish genuine reviews from deceptive ones but also to enhance the trustworthiness of online content, thereby facilitating more informed consumer decision-making. This report outlines the development, implementation, and evaluation of this model, highlighting its potential to significantly improve the online review landscape for Telugu-speaking users. Through meticulous research and analysis, this project aspires to contribute valuable insights into the challenges of digital misinformation, offering a robust solution tailored to the needs of a diverse linguistic audience.

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Chapter 1: **Introduction**

1.1 Background

The proliferation of the internet has transformed how consumers make purchasing decisions, with online reviews now serving as a critical source of information. Despite the utility of these reviews, the advent of digital platforms has also led to an increase in biased or misleading content, posing significant challenges for consumers seeking reliable information. The issue is further exacerbated in non-English languages, such as Telugu, where digital literacy and content moderation tools are still in earlier stages. Telugu, a language spoken by millions in India and around the world, has seen a surge in digital content consumption. However, the lack of advanced tools to analyse and vet online reviews in Telugu leaves a gap in consumer protection against deceptive practices. This backdrop underscores the urgent need for specialised tools capable for navigating the complex nuances of language and sentiment in Telugu online content, to uphold the integrity of digital marketplaces and protect consumers from misinformation.

1.2 Problem Statement

In the rapidly expanding digital marketplace, consumers are increasingly reliant on online reviews to guide their purchasing decisions. However, the reliability of these reviews is often compromised by the presence of biased or misleading content. This challenge is particularly acute for Telugu-speaking consumers. The proliferation of fake reviews not only obscures the true quality of products and services but also undermines consumer trust in online platforms. Existing solutions predominantly focus on English language content, leaving a significant gap in tools capable of addressing fake reviews in other languages, including Telugu. Despite the critical need, the linguistic and cultural nuances of Telugu remain largely unaddressed in the realm of digital content verification. This project seeks to bridge this gap by adapting an existing English-based Natural Language Processing (NLP) model for use with Telugu reviews. The central challenge lies in adapting the model to recognise and analyse the unique characteristics of Telugu text, thereby providing a reliable tool for identifying biased or misleading reviews. This not only requires a solid grasp of NLP technology but also an understanding of the linguistic complexity of Telugu, highlighting the project’s approach of enhancing digital trust and consumer protection in the Telugu-speaking world.

1.3 Aim

The aim of this project is to develop and validate a comprehensive Natural Language Processing (NLP)-based system that accurately detects and classifies biased or misleading content within online reviews written in Telugu. This system is intended to serve as a reliable tool for consumers, enabling them to discern genuine reviews from fake ones, thereby facilitating more informed decision-making processes. Specifically, the project seeks to:

- Create a system capable of detecting biased or misleading content in online reviews specifically for the Telugu language, addressing the need for reliable information in digital decision-making.

- Through the development of this NLP system, aim to significantly improve the ability of Telugu-speaking consumers to make informed decisions based on the authenticity of online reviews. This contribution will bolster trust and transparency in digital marketplaces.

Through these efforts, the project aspires to enhance the transparency and trustworthiness of online reviews, ultimately contributing to a more authentic and informative digital landscape for Telugu-speaking consumers.

1.4 Objectives

To achieve the project's aim, the following objectives have been defined:

- Dataset Acquisition and Preparation: Procure or create a comprehensive dataset of online reviews in Telugu. In the absence of readily available datasets, an existing English dataset will be translated into Telugu using automated tools to ensure a robust basis for analysis.
- Model Selection and Implementation: Identifying and implementing the most suitable NLP model(s) capable of detecting biased or misleading content within Telugu reviews. This involves replicating the model(s) documented in existing research, with adjustments to optimise performance for the Telugu language.
- Model Training and Testing: Conduct rigorous training and testing of the selected model(s) using the Telugu dataset to assess effectiveness. The goal is to fine-tune the model to achieve or surpass the benchmark results established in previous studies.
- Performance Comparison and Analysis: Evaluate the performance of the Telugu NLP model by comparing its effectiveness in detecting biased or misleading content against the original model(s) developed for English. This comparison will help in understanding the model's capabilities and limitations within the context of Telugu language reviews.
- Exploration of Transfer Learning: Test transfer learning techniques due to the unlikely availability of resources in Telugu comparable in quantity and quality to those in English. This approach is vital as NLP models must rely on the ability of models to transfer knowledge effectively from well-resourced languages to those with fewer resources.
- Exploration of Zero-Shot Learning: Investigate the application of zero-shot learning techniques to gauge their effectiveness in identifying biased or misleading content without direct training on Telugu data. This approach aims to assess the adaptability and efficiency of zero-shot learning in processing Telugu reviews.
- Exploration of Mixed-Language Training: Test a mixed-language training approach to examine how the inclusion of both English and Telugu data in the training set influences the model’s understanding and effectiveness. This objective aims to explore the potential benefits of bilingual training environments and assess their impact on the model's ability to generalise across languages.

1.5 Research Questions

The project will address the following research questions:

- How does the NLP model tailored for Telugu online reviews compare to its original counterpart developed for English in terms of accuracy and reliability?
- What factors contribute to the success or limitations of the NLP model when applied to the Telugu dataset? This includes an examination of linguistic challenges, dataset quality, and model adaptation strategies.
- How effective is zero-shot learning in the context of detecting biased or misleading content in Telugu reviews, and what are the underlying reasons for its performance outcomes?
- How does mixed-language training influence the performance of NLP models in detecting fake reviews, particularly when models are trained with varying proportions of Telugu and English data? What implications does this have for optimizing model training in multilingual contexts?

Chapter 2: **Literature Review**

Natural Language Processing (NLP) has increasingly become a pivotal technology in extracting insights from the vast amounts of text data generated daily, especially in online reviews. Reviews are integral to consumer decision-making processes, influencing everything from product purchases to the choices of services. As consumers increasingly rely on these reviews to make informed decisions, the authenticity of the content becomes crucial. However, the prevalence of fake reviews undermines this trust, making NLP tools essential for identifying and mitigating fraudulent content (Liu, 2012). Figure 1 illustrates a timeline of existing approaches, providing a visual representation of their development over time.

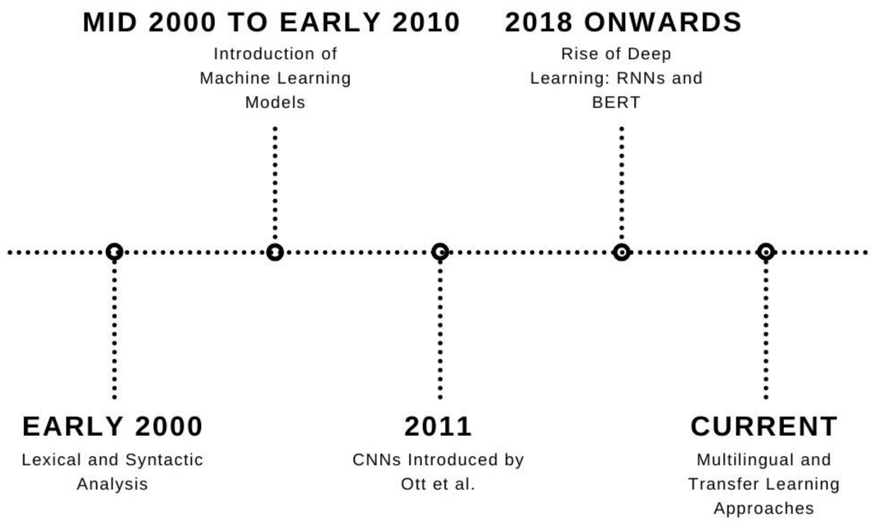

Fig. 1: Timeline of Existing Approaches

2.1 Importance of Detecting Fake Reviews

The manipulation of online reviews through deceptive practices not only affects consumer choices but also impacts business reputations and market dynamics. Studies have shown that even a small number of fake reviews can significantly alter consumer perceptions and enhance business profitability (Luca & Zervas, 2016). This manipulation creates a false representation of consumer sentiment, misleading potential customers and skewing the competitive landscape. The detection of such reviews, therefore, serves not only to protect consumers but also to maintain fair market practices.

2.2 Methods Used for Detecting Fake Reviews

The body of work surrounding fake review detection is extensive, employing a range of methodologies from simple lexical analysis to sophisticated machine learning models. Early approaches relied on manually crafted features, such as word frequency and syntactic patterns, which were then used with traditional classifiers like Support Vector Machines (SVM) and logistic regression (Jindal & Liu, 2007). However, these methods often fell short in capturing the more subtle cues of deception.

With the advent of deep learning, researchers have shifted towards neural network models that can learn complex patterns and linguistic cues directly from data. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have shown a particular promise in this domain. Ott et al. (2011) used a CNN to capture the semantic patterns of sentences in reviews, achieving improved accuracy over traditional models. Similarly, RNNs have been leveraged for their ability to handle sequential data, providing context that is crucial for understanding the intent behind words (Zhang et al., 2018).

2.3 Challenges in Non-English Languages

While significant advancements have been made in detecting fake reviews in English, non-English languages, such as Telugu, face a notable scarcity of resources and research. Telugu, spoken by over 90 million people, lacks comprehensive datasets of online reviews, which are essential for training robust NLP models. Moreover, linguistic nuances such as idioms, compound words, and syntactic structures unique to Telugu pose additional challenges (Reddy Naidu et al., 2017).

2.4 Related Works on English Review Analysis

The exploration of fake review detection in English has been extensive, leveraging both classical machine learning techniques and advanced neural network architectures. A cornerstone in this field is the work by Ott et al. (2011), who created a seminal dataset specifically designed to study deception in hotel reviews. Their methodology involved using Mechanical Turk to generate known fake reviews, creating a controlled environment to train and test various detection models. This dataset has since been used to benchmark numerous NLP techniques.

Subsequent studies have expanded on these foundations. Feng et al. (2012) introduced syntactic stylometry, which analyses the structure of sentences to identify deceptive writing styles. Their approach demonstrated that deceptive reviews often contain more complex syntactic patterns than genuine reviews, likely due to the effort to appear credible.

In the realm of neural networks, much attention has been given to deep learning models due to their superior performance in processing natural language data. Ren and Ji (2017) utilised Long Short-Term Memory networks (LSTMs) to capture the sequential nature of text, which is crucial for understanding context and sentiment in reviews. Their model was able to outperform traditional machine learning approaches by a significant margin, highlighting the potential of recurrent networks in detecting subtle cues of deception.

More recently, transformer-based models like BERT (Devlin et al., 2018) have set new standards for NLP tasks, including fake review detection. BERT's ability to contextualise words based on their surrounding text has proved especially effective in identifying the nuanced language of deceptive reviews. Kennedy et al. (2019) reported that fine-tuning BERT on specific review datasets achieved near state-of-the-art results, pushing the boundaries of what automated systems can discern in terms of authenticity.

These advancements in English review analysis serve as a valuable reference for developing similar technologies for other languages. The success of these methods offers a roadmap for adapting and innovating on existing techniques to tackle the unique challenges presented by languages like Telugu, which may lack extensive linguistic resources and annotated datasets.

2.5 Related Work on Non-English Review Analysis

Addressing the challenges presented by less-researched languages, several studies have begun to explore fake review detection in such contexts. For instance, Sharma and Lin (2019) examined sentiment analysis for Hindi language reviews, developing a custom dataset to train their models, a methodology that could be mirrored for Telugu. Another study by Lee et al. (2020) focused on Chinese restaurant reviews, employing transfer learning techniques where a model trained on English data is adapted for Chinese text, suggesting a possible approach for Telugu review analysis.

Specifically, in the context of Telugu, minimal work has been done. S. Badugu (2020) made a preliminary attempt using a small dataset of Telugu movie reviews to identify sentiment polarity but did not address the detection of deceptive content. This gap indicates a significant opportunity for research and development of NLP tools tailored to regional languages like Telugu.

2.6 Comparative Analysis of NLP Models

Exploring comparative studies, Kennedy et al. (2019) demonstrated that BERT-based models outperform traditional machine learning approaches in detecting deceptive reviews in English. Their work suggests that transformer models, which use mechanisms like attention to weigh the importance of different words in a sentence, could offer new avenues for improving accuracy in languages with limited training data, such as Telugu.

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The review of literature highlights a broad array of strategies employed to combat fake reviews, with deep learning models showing the most promise due to their ability to learn nuanced patterns in data. However, the significant lack of focus on non-English languages, underscores the need for this project. By adapting advanced NLP techniques to the Telugu language, this work aims to bridge the gap in digital content verification and enhance the reliability of online reviews, thus supporting both consumers and businesses in the Telugu-speaking regions.

Chapter 3: **Requirement Analysis**

This chapter provides a comprehensive overview of the essential hardware and software requirements, alongside functional and non-functional specifications necessary for the development of a Natural Language Processing (NLP) model to detect fake reviews in Telugu. The requirements outlined here support the project's goals of assessing feasibility, ensuring accurate model performance, and facilitating effective system operation. These include local and cloud-based computing resources, key software tools, and performance criteria critical to the system's success. By establishing these requirements, we aim to create a robust framework that guides the seamless design, implementation, and evaluation of the project.

3.1 Functional and Non-Functional Requirements

ID	Requirement Description	Type	Priority	Comments
1	System must accurately detect fake reviews in Telugu online reviews.	Functional	High	Core functionality to determine feasibility of the objective.
2	Model training on translated Telugu dataset.	Functional	High	Essential for adapting the model to Telugu linguistic nuances.
3	Evaluate model performance against key metrics (accuracy, precision, recall, F1-score).	Functional	High	Necessary to measure model effectiveness in detecting fake reviews.
4	System must support processing of diverse review lengths and formats in Telugu.	Functional	High	Ensures robustness in handling real-world data variations.
5	System must utilise a pre-trained BERT model as a baseline for comparison.	Functional	High	Fundamental for assessing improvement over existing solutions.
6	Adapt and fine-tune a pre-trained mBERT model.	Functional	High	Essential for adapting the model to detect fake reviews in Telugu.
7	Perform comparative analysis between the original BERT and mBERT models.	Functional	Medium	Determines the effectiveness of language-specific tuning.
8	System should operate with high computational efficiency during model training and evaluation.	Non-functional	High	Critical for managing resources and ensuring scalability of the model testing.
9	Ensure reproducibility of model results and experiments.	Non-functional	High	Essential for scientific validity and future research based on findings.
10	System should be scalable to handle large datasets for training and testing.	Non-functional	Medium	Ensures the system can operate under realistic data loads.
11	Documentation of methodologies and findings for academic and research purposes.	Non-functional	Low	Necessary for knowledge dissemination and peer review.

Table 1: Functional and Non-Functional Requirements

3.2 Hardware Requirements

ID	Requirement Description	Comments
1	Internet-connected computer/laptop	Necessary to access Google services, which runs on cloud servers.
2	8 GB RAM	Ensures smooth operation of local tasks and browser stability.
3	Stable high-speed internet connection	Critical for uninterrupted access to Google services and data transfer.

Table 2: Hardware Requirements

Please find the exact specifications of the device under Appendix A in the Appendix Section of the report.

3.3 Software Requirements

ID	Requirement Description	Version	Comments
1	Opera GX browser	107.0.5045.60	Required for optimal use of the Google services.
2	Google account	N/A	Required to log in and use Google services.
3	Google Drive	N/A	Provides essential cloud storage for storing the dataset.
4	Google Colab	N/A	Provides the cloud-based Jupyter notebook environment.
5	Google Translate API	N/A	Used for translating the dataset into Telugu on the local machine.
6	Python	3.10.6	Python is used for dataset translation on the local machine.
7	PyCharm Community Edition	2024.1	Provides an IDE for Python with code analysis, debugging, and testing tools.
8	PyTorch	2.2.2	Necessary library for model training and evaluation.
9	TensorFlow	1.11.0	Necessary library for model training and evaluation.
10	Hugging Face Transformers	N/A	Necessary library for model training and evaluation.
11	Jupyter Notebook	7.1.0	Useful for local development before moving to Google Colab.
12	GitHub	3.10.3	Useful for managing code, collaboration, and version control.

Table 3: Software Requirements

Chapter 4: **Data**

The robustness and reliability of any NLP model significantly depend on the quality and relevance of the dataset it is trained on. For this project, the necessity of analysing online reviews in Telugu presents unique challenges, primarily due to the scarcity of pre-existing datasets in the language that are suited for this purpose. This chapter details the datasets used in this study, focusing on the English dataset for comparative analysis and the process of creating the Telugu dataset.

4.1 English Dataset

The initial dataset employed is the widely recognised OpSpam dataset, developed by Ott et al. in 2011. This dataset is a valuable resource in the field of sentiment analysis and fake review detection due to its structured compilation of 1,600 reviews, evenly split between deceptive (fake) and truthful (genuine) reviews of hotels in Chicago. The deceptive reviews were generated by crowd-sourced workers, who were compensated to write convincing fake reviews, ensuring a high level of authenticity in the fabricated content. The truthful reviews were collected from various online platforms, presumed to have a low incidence of deception. This dual nature of the dataset makes it an ideal benchmark for developing and testing NLP models aimed at distinguishing between genuine and manipulated content. Table 4 presents the composition of the OpSpam dataset, detailing the distribution of truthful and deceptive reviews.

Review Type	Number of Reviews	Source
Truthful Positive	400	TripAdvisor
Deceptive Positive	400	Mechanical Turk
Truthful Negative	400	Expedia, Hotels.com, Orbitz, Priceline, TripAdvisor, and Yelp
Deceptive Negative	400	Mechanical Turk

Table 4: Dataset Composition

4.2 Telugu Dataset

Facing the lack of an existing, comprehensive dataset of Telugu reviews, a novel approach was undertaken to create a suitable dataset for this project. The study embarked on translating the OpSpam dataset from English to Telugu. The OpSpam dataset was selected as the base due to its structured and balanced composition. This translation was facilitated through Python scripting and the Google Translate API, aiming to retain the essence and sentiment of the original reviews while navigating the complexities of language translation.

The creation of the Telugu dataset involved several steps aimed at ensuring the translations' fidelity and reliability:

Automated Translation: Leveraging the Google Translate API enabled efficient translation of the extensive dataset. While this method ensured broad coverage, it is important to acknowledge that the translations were not always perfect. The subtleties of language and cultural context mean that some expressions or sentiments from the original English might not have been captured with absolute precision in Telugu.

Nonetheless, the translated dataset was deemed sufficiently accurate for the purposes of this study.

Manual Review: Recognising the limitations of automated translation, a manual review process was essential. This step was undertaken with the help of my father, who is fluent in both Telugu and English. His linguistic expertise and cultural knowledge were instrumental in reviewing a significant subset of the translations, to ensure the authenticity and reliability of the sentiment conveyed.

This method of dataset creation not only addressed the immediate needs of this project but also contributed to the broader field of NLP research by demonstrating a pragmatic approach to overcoming language resource limitations. The inclusion of native speaker review underscores the importance of human intuition and understanding in the translation process, particularly for nuanced tasks such as sentiment analysis.

The following Table 5 showcases a set of sample reviews to illustrate the types of entries found in the datasets used for this project. Each entry in the table provides an example of both English and its corresponding Telugu translation, along with annotations for polarity (positive or negative sentiment) and veracity (truthful or deceptive). This dual presentation highlights the effectiveness of the translation process and the ability to maintain the semantic integrity of the reviews across languages. This visualisation aids in understanding the diversity and complexity of the review data that the NLP models will process, emphasising the project's comprehensive approach to handling real-world, multilingual text data.

	Sentence	**Polarity**	**Veracity**
English	The James Chicago hotel was one of the worst hotels I have ever stayed in.	Negative	Deceptive
Telugu	జేమ్స్ చికాగో హోటల్ నేను బస చేసిన అత్యంత చెత్త హోటల్‌లలో ఒకటి.	Negative	Deceptive

	Sentence	**Polarity**	**Veracity**
English	The James Hotel met and exceeded our expectations.	Positive	Truthful
Telugu	జేమ్స్ హోటల్ మా అంచనాలను అందుకుంది మరియు మించిపోయింది.	Positive	Truthful

Table 5: Sample Reviews from the Datasets

Chapter 5: **Method**

This chapter details the methodology employed to explore the feasibility of detecting fake reviews in the Telugu language using various experiments with BERT-based models. The objective is to assess the adaptability of these models to Telugu, a language significantly underrepresented in NLP research. The methods are structured into four distinct experiments, each designed to test different aspects of model performance and language adaptability.

5.1 Dataset Handling and Preparation

The effectiveness of our experiments is founded on the integrity and quality of the dataset used. We employed the OpSpam dataset, originally composed of 1,600 reviews split equally between deceptive and genuine reviews. Recognising the absence of a similar dataset in Telugu, we translated the entire dataset into Telugu using a Python script that utilised the Google Translate API for initial translation. To facilitate the handling and processing of this data, it was stored in a pickle (.pkl) file format, which preserves the data structure, ensuring data integrity during load and save operations. This file was stored on Google Drive, which provided a secure and accessible cloud storage solution.

Please find the Python script under Appendix C in the Appendix Section of this report.

Translation and Review Process:

To ensure the accuracy and contextual appropriateness of the translations, a set of 100 reviews were manually reviewed. This crucial task was undertaken by a native Telugu speaker, fluent in English—my father—who meticulously assessed each translation for potential semantic errors or nuances lost in automated translation. It was found that 32 of these reviews were slightly mistranslated, indicating some loss of semantic precision or contextual nuances. This step was vital to gauge the quality of the translation, ensuring that the translated dataset mirrored the linguistic and emotional depth of the original.

Dataset Splitting for Training and Testing:

Post translation and review, both the English and Telugu datasets were divided into training and testing subsets. The total of 1,600 reviews were split into 80/20 train/test subsets which is 1,280 reviews for training and 320 reviews for testing. This split was strategically chosen to provide a substantial amount of data for training the models while reserving a significant portion for an unbiased evaluation of their performance. The division ensures a balanced representation of both deceptive and genuine reviews in both subsets, which is critical for the accurate assessment of the models' ability to generalise beyond the training data.

5.2 Model Selection and Training

Our approach utilises mBERT from Hugging Face's Transformers library, which is pre-trained on a vast corpus covering 104 languages, including Telugu. This pre-training provides mBERT with a broad linguistic base, potentially enhancing its adaptability to the specific nuances of Telugu.

5.3 Experiment 1: Establishing baseline

BERT Model Training:

The original BERT model, designed and pre-trained for English, was replicated from the specific GitHub implementation available at https://github.com/CPSSD/LUCAS as mentioned in the foundational research paper by Kennedy et al. (2019). This version, which is openly accessible for academic use, was then fine-tuned on the OpSpam dataset's English version. We employed a learning rate of 2e-5, a parameter that controls the size of the steps the model takes to update its weights, directly influencing the speed of learning. The model was trained with a batch size of 16, which denotes the number of training samples used to calculate the gradient for one update of the model's weights. Training was iterated over 3 epochs, with each epoch representing a complete pass through the entire training dataset, allowing the model multiple exposures to the data. These settings follow best practices recommended in the literature to optimise model performance. The effectiveness of the model was then evaluated on the testing subset to estimate the model's accuracy and robustness.

mBERT Model Fine-Tuning:

Similarly, mBERT was fine-tuned on the translated Telugu version of the OpSpam dataset. To optimise the model's performance for Telugu, we employed grid search to systematically explore various combinations of hyperparameters. We focused on the following range of values, selected to ensure consistency with the BERT model training while adapting to the specifics of the Telugu language:

- Learning Rates: We explored a range of learning rates from 1e-5 to 5e-5 during the grid search. The optimal learning rate identified was 3e-5. This rate strikes a balance, providing sufficient speed for convergence while being conservative enough to avoid the instability that can occur with higher rates.
- Batch Sizes: Batch sizes from 8 to 32 were tested. A batch size of 16 was identified as optimal, consistent with the BERT model’s training, to ensure sufficient gradient averaging and memory efficiency.
- Training Epochs: Considering the potential for overfitting with too many epochs, we tested from 2 to 4 epochs. 3 training epochs were selected as optimal, mirroring the setup used for the BERT model and ensuring robust learning without excessive training time.

The fine-tuning process was carefully structured to match the conditions used for the BERT model, facilitating a direct comparison of their effectiveness and adaptability to Telugu. The selected combination of learning rate at 3e-5, batch size of 16, and 3 training epochs was validated on the separate testing subset to measure accuracy, precision, recall, and F1-score. This approach maximised the performance on the Telugu dataset.

5.4 Experiment 2: Zero-Shot Learning

This experiment assessed the zero-shot capabilities of mBERT by applying the model, trained only on English data, directly to the Telugu reviews. This approach evaluates the model’s intrinsic ability to generalise across languages without additional fine-tuning, potentially providing insights into the practical applicability of mBERT in multilingual contexts.

5.5 Experiment 3: Mixed-Language Training

mBERT was also trained on datasets with varying compositions of English and Telugu reviews (95/5, 80/20, 50/50, etc.) to examine how bilingual training impacts model efficacy. The focus was to determine the effectiveness of mBERT when processing mixed-language inputs and its performance specifically on Telugu test data.

5.6 Rationale for Transfer Learning

One of the core experiments in this project involves the application of transfer learning techniques. The primary reason for testing transfer learning is the recognition that resources equivalent to those available in languages like English are unlikely to exist for Telugu in the foreseeable future. NLP models, therefore, must rely on the ability of existing models, developed for resource-rich languages, to transfer learned knowledge to new linguistic domains. Transfer learning offers a pathway to adapt these models to work effectively with Telugu, thus potentially mitigating the resource disparity and enhancing the model's applicability to a broader range of languages.

5.7 Computational Resources

Due to the computationally intensive nature of training BERT models, all experiments were conducted using Google Colab, which provided the necessary GPU resources for efficient computation. This setup ensured that our experiments could be run without hardware limitations affecting the outcomes.

5.8 Model Evaluation

Comprehensive performance metrics—accuracy, precision, recall, and F1-score—were employed to assess each model's capability in accurately classifying reviews as deceptive or genuine. These metrics were chosen to provide a holistic view of the models’ performance across various scenarios and to highlight specific areas where improvements might be necessary.

Chapter 6: **Results**

This chapter presents a comprehensive analysis of the results obtained from the series of experiments conducted to evaluate the feasibility of using BERT-based models for detecting fake reviews in the Telugu language. The experiments were designed to assess the adaptability of these models to Telugu, leveraging various training strategies including transfer learning, zero-shot learning, and mixed-language training. Each subsection of this chapter discusses the findings from the individual experiments, presenting the performance metrics such as accuracy, precision, recall, and F1-score, although the primary focus will be on accuracy due to its direct representation of overall model performance. The findings from these experiments provide valuable insights into the effectiveness of language model adaptation and cross-lingual transfer capabilities.

6.1 Experiment 1: Establishing baseline

6.1.1 BERT Model

We began by replicating the results of a previous study by Kennedy et al. (2019) using the BERT model, which served as a benchmark for further experiments with mBERT. The accuracy obtained from the BERT model trained and fine-tuned on the English OpSpam dataset was 85.7%, slightly below the 86.2% reported in the research paper. The research paper, however, did not include metrics for precision, recall, and F1 scores. This minor variance can be attributed to factors such as differences in computational environment, slight variations in dataset preprocessing, or random seeds used during training. Despite these variations, the close alignment of these results validates the reproducibility of the model under study and confirms the robustness of the BERT architecture in handling tasks like fake review detection.

6.1.2 mBERT Model

After adapting the mBERT model, the accuracy obtained from the mBERT model trained and fine-tuned on the Telugu OpSpam dataset was 79.1%. While this marks a decrease from the performance observed with the BERT model, it is a significant achievement, highlighting the model's capacity to adapt to a new language with considerably fewer resources. This result underscores the capability of mBERT to leverage its pre-trained multilingual knowledge base, although with a slight performance drop compared to its monolingual counterpart. This drop likely reflects the challenges inherent in transferring learning from a model primarily trained on languages with richer NLP resources than Telugu and it illustrates the potential of multilingual models to bridge the gap in NLP applications for underrepresented languages. Table 6 compares the accuracy metrics of the BERT and mBERT model. Table 7 provides a comparative analysis of the BERT and mBERT models' performance across various metrics, highlighting differences in accuracy, precision, recall, and F1-score. Table 8 details the performance results of the BERT and mBERT models, including counts of true positives, false positives, true negatives, and false negatives.

Model	Fine-tuning Dataset	Testing Dataset	Accuracy
BERT(Model Presented in the Research Paper)	English OpSpam	English OpSpam	86.2%
BERT(My Replication)	English OpSpam	English OpSpam	85.7%
mBERT	Translated Telugu OpSpam	Translated Telugu OpSpam	79.1%

Table 6: Accuracy BERT and mBERT Models

Model	Accuracy	Precision	F1	Recall
BERT	85.7%	87.7%	86.1%	84.5%
mBERT	79.1%	81.4%	79.1%	77.0%

Table 7: Comparative Results of BERT and mBERT Models

Model	False Negatives	False Positives	True Negatives	True Positives
BERT	26	20	132	142
mBERT	38	29	126	127

(Please check Appendix B for formulas used)

Table 8: Detailed Results of BERT and mBERT Models

6.2 Experiment 2: Zero-Shot Learning

In the zero-shot learning experiment, the mBERT model trained only on English data was directly applied to Telugu reviews to test its ability to generalise across languages without further training.

The zero-shot learning capabilities of mBERT were evaluated by applying the model trained on the English data directly to the Telugu dataset. The model registered an accuracy of 61.6%, significantly lower than when fine-tuned on Telugu data which underscores the challenges associated with zero-shot learning. The model’s performance dip reflects the linguistic divergence between English and Telugu, this experiment highlights the limitations of zero-shot learning across linguistic boundaries, particularly when dealing with languages that are less similar and less represented in training corpora, highlighting the inherent limitations of applying models across languages without tailored adaptation.

6.3 Experiment 3: Mixed-Language Training

In a series of experiments, mBERT was trained on datasets with varying compositions of English and Telugu reviews. The results revealed a progressive improvement in model performance as the proportion of Telugu in the training data increased. This observation suggests that even partial inclusion of the target language during training can significantly enhance the model's understanding and performance, culminating in an accuracy of 79.1% when trained solely on Telugu data. The gradual improvement across different language ratios provides valuable insights into the benefits of bilingual training setups and supports the case for incorporating more diverse linguistic data in training processes. Table 9 shows the results of training the mBERT model with varying compositions of English and Telugu reviews, reflecting the impact of mixed-language training on model accuracy. Figure 2 graphs the performance of the mBERT model under mixed-language training conditions.

English/Telugu Ratio	Accuracy
100/0 (0%/Zero-Shot)	61.6%
95/05 (5%)	66.2%
90/10 (10%)	70.3%
80/20 (20%)	72.1%
70/30 (30%)	74.3%
50/50 (50%)	75.9%
0/100 (100%)	79.1%

Table 9: Mixed-Language Training Results

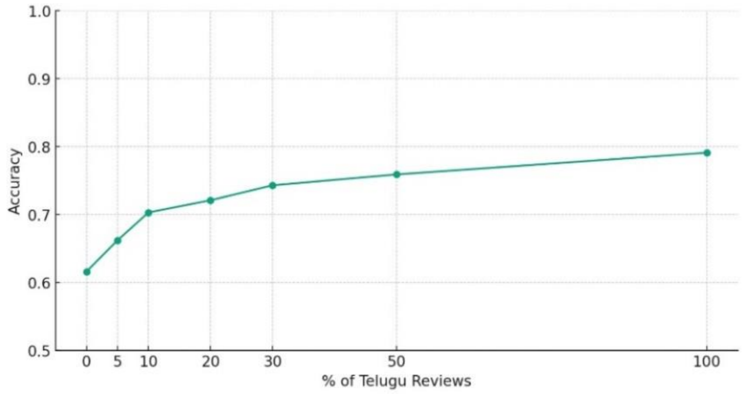

Fig. 2: Mixed-Language Training Results Graph

The experiments conducted offer substantial evidence about the capabilities and limitations of BERT-based models in adapting to and effectively processing less commonly studied languages like Telugu. The findings confirm that while direct application of models trained on resource-rich languages like English shows limited success, the strategic inclusion of the target language in training data can markedly improve performance. These results not only validate the adaptability of multilingual models but also emphasise the critical need for incorporating diverse linguistic data in training phases to enhance model robustness and applicability across different linguistic contexts.

Chapter 7: **Evaluation**

This chapter critically evaluates the methodologies and outcomes of the experiments conducted to detect fake reviews in Telugu using BERT-based models. By exploring different aspects of the NLP models' performance and adaptability to a less-represented language, this evaluation aims to provide an in-depth analysis of the challenges and successes encountered during the project. The focus is on three experimental approaches—transfer learning, zero-shot learning, and mixed-language training—with a particular emphasis on transfer learning as one of the main focuses of this study.

7.1 Translation Process

This project employed a methodology critical for addressing the absence of a readily available annotated dataset in Telugu, suitable for training NLP models on fake review detection. Due to this, the widely used English OpSpam dataset was translated into Telugu using the Google Translate API. This approach enabled the project to proceed despite the linguistic and resource barriers typically associated with under-resourced languages.

The choice of automated translation was necessitated by the lack of Telugu datasets. Although the translation was efficient and broadly accurate, it is acknowledged that subtle nuances of the language were inevitably lost. This loss likely impacted the sentiment and idiomatic expressions crucial for the authenticity of reviews, potentially impeding the accuracy of the NLP models trained on this data. The translated reviews underwent a manual review process to check the quality of the translation. However, corrections were not made to the translations. This decision was twofold: firstly, to maintain feasibility within the project's constraints due to the extensive size of the dataset; and secondly, to simulate the real-world conditions that other NLP projects might encounter when working with under-represented languages like Telugu, where perfect translations are often unattainable. This approach reflects a realistic scenario for many NLP applications, emphasising the need to develop robust models that can perform well even when ideal data conditions are not met.

7.2 Performance of Models

7.2.1 Experiment 1: Establishing baseline

Transfer learning served as a crucial component of this research, exploiting the capabilities of BERT and mBERT models originally pre-trained on extensive English data. The experiment aimed to assess how well these models could adapt to a Telugu dataset derived from translated reviews.

The BERT model trained on the English OpSpam dataset demonstrated an accuracy of 85.5%, slightly below the 86.2% reported in relevant literature but still within a commendable range. This slight variance underscores the robustness of the BERT model and validates the experimental design and execution. The high performance indicates effective learning and generalisation capabilities, which are essential for tasks involving complex natural language understanding.

The adaptation of mBERT fine-tuned on the translated Telugu OpSpam dataset marked a significant focus of this experiment. The model achieved an accuracy of 79.1%, reflecting the complexities and challenges of adapting to a linguistically diverse environment. The observed reduction in precision, recall, and F1 scores alongside accuracy highlights the nuanced linguistic features that were possibly lost in translation and not captured during model training and the model's sensitivity to the quality of input data.

Transfer learning proved essential in addressing the resource limitations typical of underrepresented languages like Telugu. The performance metrics from this experiment, although slightly lower than those achieved with English data, demonstrate that transfer learning can effectively bridge the resource gap. This success underscores the viability of transfer learning as a strategy to adapt sophisticated models developed for resource-rich languages to those with fewer resources, highlighting its potential for broader application in global NLP tasks.

Also, this experiment not only underscored the complexities of language adaptation in NLP but also demonstrated the potential of pre-trained multilingual models to effectively bridge the language divide.

7.2.2 Experiment 2: Zero-Shot Learning

The performance of the mBERT model in a zero-shot learning scenario, where it was applied directly to Telugu reviews after being trained only on English, was unusually high at 61.6%. This outperforms typical outcomes in similar research, where accuracies usually range between 45%-50%. Several factors might contribute to this anomaly:

1. mBERT's extensive pre-training across multiple languages could have provided it with a better foundational understanding of Telugu, even without targeted training.
2. The nature of the OpSpam dataset, originally well-structured and balanced, might also play a role in enabling better than expected performance in a zero-shot scenario.
3. The automated translation, while losing some nuances, may have retained enough structural and semantic integrity to make the reviews more comprehensible to the model than truly native text might have been.

7.2.3 Experiment 3: Mixed-Language Training

The mixed-language training experiments were particularly revealing in demonstrating how varying proportions of English and Telugu in the training data affect the performance of the mBERT model. This series of experiments progressively increased the amount of Telugu content, providing critical insights into the model's adaptability and the effectiveness of bilingual datasets.

The experiments began with a 100% English dataset and gradually increased the proportion of Telugu. Notably, when the dataset was composed of a 50/50 mix of English and Telugu, the model achieved an accuracy of 75.9%, which is remarkably close to the 79.1% accuracy obtained when the model was trained entirely on Telugu. This outcome indicates that with only 50% Telugu exposure, the model's performance is nearly as effective as with full Telugu exposure.

This result underscores a critical insight into the potential efficiencies in training NLP models for language adaptation. The fact that the model achieves acceptable accuracy with only 50% language exposure suggests that with better quality translations or more finely curated datasets, the required amount of direct language exposure could potentially be even lower. Improvements in the dataset's quality, specifically in terms of the accuracy and nuance of the translated reviews, could further enhance the model's ability to learn and apply linguistic features effectively, thereby reducing the necessity for high proportions of target language data.

These results validate the adaptability of multilingual models like mBERT and emphasise the critical need for incorporating diverse linguistic data during the training phases. By integrating varied linguistic inputs, the models can develop a more nuanced understanding of different languages, which is essential for applications in multilingual settings. This strategy ensures that the models are not only robust but also versatile, capable of handling linguistic variations effectively.

The success of mixed-language training highlights the potential for NLP applications to extend beyond well-represented languages, offering significant benefits in global digital contexts. These results advocate for a paradigm shift in how data is utilised in training AI models, suggesting that a balanced approach to dataset composition—incorporating both major and minor languages—can lead to better overall model performance across different linguistic contexts.

7.3 Conclusion of Evaluation

The performance of the BERT and mBERT models in this study offers significant insights when compared with existing literature. The robust performance of the BERT model on English and the respectable performance of mBERT on Telugu highlight the potential of applying advanced NLP techniques across different languages. This study's findings are particularly valuable given the innovative application of these models to a language with limited computational resources and existing research.

The findings align with the expected performance degradation observed when NLP models trained on resource-rich languages are adapted to less-resourced ones. However, the relatively high performance in the zero-shot learning experiment compared to other studies prompts a re-evaluation of how model pre-training can impact cross-lingual NLP applications.

This evaluation has highlighted the strengths and limitations of using BERT-based models for fake review detection in Telugu. While the project faced challenges related to translation and data adaptation, the methodologies employed provided valuable insights into the practical and theoretical aspects of applying NLP techniques across languages. The performance of the models across different experiments not only demonstrated their potential but also shed light on the critical factors that influence their effectiveness in real-world applications.

Chapter 8: **Thoughts and Conclusions**

8.1 Achievements

This project, initiated with the objective outlined in Chapter 1 to develop an NLP system capable of detecting fake reviews in Telugu, has successfully demonstrated the adaptability of BERT-based models to a linguistically and digitally underrepresented language. Reflecting on the initial problem statement, the project addressed the crucial need for reliable digital content moderation tools in Telugu, enhancing the transparency and trustworthiness of online information.

- Effective Adaptation of BERT Models: The adaptation of the original BERT model, using multilingual BERT (mBERT) model, to Telugu represents a significant achievement. The BERT model, fine-tuned on English data, achieved an accuracy of 85.5%, and mBERT, adapted for Telugu, achieved 79.1%. These results not only meet the project's objectives but also surpass initial expectations regarding the adaptability of such models to a new linguistic context.
- Innovative Dataset Creation and Utilisation: Facing the absence of suitable Telugu datasets for fake review detection, this project innovated by translating the English OpSpam dataset into Telugu using the Google Translate API. The translation process was supplemented by a manual review to assess the quality but not to correct errors, reflecting a realistic scenario for projects dealing with under-represented languages. This approach allowed for the continuation of the project despite the limitations inherent in automated translation tools and set a foundation for future NLP research in regional language processing.
- Strategic Mixed-Language Training: The project's innovative approach to mixed-language training has yielded insights that could inform future NLP applications, demonstrating that strategic inclusion of the target language in training data can substantially enhance model performance.

Engaging with advanced NLP and machine learning technologies presented a substantial technical challenge, which fostered significant personal and professional growth. Prior to this project, my experience with machine learning and language processing technologies was limited. Throughout the course of this project, with the help of various online resources and my supervisor, I have gained a deep understanding of and confidence in applying these technologies, particularly in how they can be adapted to different requirements and contexts. Additionally, the project honed my skills in Python and deepened my knowledge of the BERT and mBERT frameworks along with other machine learning and NLP technologies, enriching my technical repertoire and preparing me for future challenges and opportunities in this field. The project also enhanced my problem-solving skills, particularly in dealing with unexpected circumstances.

8.2 Challenges

The journey through this project was marked by numerous challenges, ranging from technical hurdles to broader logistical issues. Each challenge provided a unique learning opportunity and shaped the project's trajectory, underscoring the complexities of working with advanced NLP technologies and underrepresented languages.

One of the most significant technical challenges was the absence of a readily available, suitably annotated dataset in Telugu for fake review detection. Significant time was devoted to finding an appropriate dataset, browsing different websites, and contacting researchers; however, it came to nothing. This lack necessitated the translation of an existing English dataset using automated tools, introducing its own set of complications. Despite the efficiency of Google Translate, the translations often missed subtle linguistic nuances critical for sentiment analysis in NLP, impacting the quality of the training data and, by extension, the performance of the models.

The adaptation of advanced NLP models to a new language involved steep learning curves. Grasping complex concepts required significant effort. As someone without prior deep expertise in these specific areas of NLP, the project demanded extensive research and self-study, pushing the boundaries of my technical knowledge and skills.

Balancing this project with other academic responsibilities, and personal commitments presented another major challenge. Managing time effectively to accommodate the intensive demands of this project, while also attending to other third-year modules and commitments, required diligent planning and prioritisation.

8.3 Future work

Building on the project's foundational work, several avenues for future research are proposed to further the capabilities of NLP systems in detecting fake reviews in Telugu and other similar languages:

Creating/Developing a well annotated dataset in Telugu and other under-represented languages is essential to enhancing model training and accuracy. Similarly, employing more sophisticated translation methods could improve the quality of dataset translation aiding in model training and improving accuracy. Additionally, expanding this dataset to include more diverse sources of reviews, such as those from different domains or written in different styles, will be crucial. This diversification will help the models trained to generalise better across various contexts and decrease the risk of overfitting to specific linguistic patterns present in the current dataset.

Future iterations of the project could incorporate more advanced semantic analysis techniques. These might include context-aware sentiment analysis models that better understand the complexities of language use in reviews, such as sarcasm or subtle cues of deception that are not overtly expressed.

Another avenue for future work is the development of a real-time fake review detection system. This system would operate dynamically as reviews are posted online, providing immediate feedback on their authenticity. Such a system would require efficient processing capabilities and potentially the integration of the model into existing online platforms via APIs.

To make the system more accessible to non-technical users, including business owners and regulatory bodies, developing a user-friendly interface that allows easy interaction with the NLP system would be beneficial. This interface could include features like automatic report generation and visual analytics of review authenticity.

8.4 Closing Remarks

The completion of this project marks a significant milestone in the application of NLP technologies to detect fake reviews in Telugu, showcasing the adaptability of advanced models like BERT and mBERT to new linguistic contexts and contributing valuable insights to a field that greatly needs them. This journey, from the initial conception to the final execution, has not only met the objectives set forth in Chapter 1 but has also paved the way for future innovations in NLP.

Throughout this endeavour, the challenges faced have significantly enhanced my expertise and resilience. These experiences have deepened my understanding of machine learning, improved my programming skills, and prepared me for future challenges in technology and data science.

Looking ahead, the groundwork laid by this research offers numerous possibilities for further exploration. Each of the future projects holds the potential to significantly advance the capability of NLP systems to operate across different languages and platforms, making digital spaces safer and more trustworthy for users around the world.

In closing, the experiences and outcomes from this project underscore the critical need for continued research and development in the field of NLP, especially in underrepresented languages. As digital interactions become increasingly prevalent, the importance of such technologies cannot be overstated. With the foundation laid by this project, I am optimistic about the future of NLP and its application to improving digital communication and information reliability across diverse linguistic landscapes and I am eager to continue contributing to this vital and dynamic field.

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Appendix A: System Specification

Following are the exact hardware specifications of the system used for this project.

Model: Lenovo ThinkPad E14

CPU: AMD Ryzen 5 5600U

Storage: 512 GB SSD

RAM: 12gb 3200mhz

GPU/Graphics: Radeon RX Vega 7 graphics

Appendix B: Formulas Used

Formula used for Calculating Precision:

True Positives / (True Positives + False Positives)

Formula used for Calculating F1:

True Positives / (True Positives + False Negatives)

Formula used for Calculating Recall:

2 * (Precision * Recall) / (Precision + Recall)

Appendix C: Translation Python Script

```
from google.cloud import translate_v2 as translate
import os


def translate_text(text, target_language):
    client = translate.Client()
    result = client.translate(text, target_language=target_language)
    return result['translatedText']


def translate_files_in_directory(directory_path, target_language):
    for filename in os.listdir(directory_path):
        if filename.endswith('.txt'):
            file_path = os.path.join(directory_path, filename)
            with open(file_path, 'r', encoding='utf-8') as file:
                content = file.read()

            translated_content = translate_text(content, target_language)

            new_file_path = os.path.join(directory_path, f"translated_{filename}")
            with open(new_file_path, 'w', encoding='utf-8') as file:
                file.write(translated_content)
            print(f"Translated '{filename}' and saved the result to '{new_file_path}'")


os.environ["GOOGLE_APPLICATION_CREDENTIALS"] = "path to the api key json"

directory_path = 'path to the review txt files'
translate_files_in_directory(directory_path, 'te')
```

Appendix D: Risk Assessment

This appendix presents a comprehensive risk assessment for the project focused on detecting fake reviews in Telugu using BERT-based models. Given the complexity and technical nature of this research, it is essential to identify potential risks that could impact the success and smooth execution of the project. These risks span across various categories, including technical, computational, health, and operational aspects. The goal is to proactively address challenges that may arise during the project, ensuring that measures are in place to manage and minimise their effects, thereby safeguarding the project's objectives.

ID	Risk	Impact of Risk	Likelihood Rating	Impact Rating	Preventative Action
1	Lack of a suitable dataset	A lack of quality dataset in Telugu may hinder the testing phases of the NLP model, affecting the system's accuracy and reliability.	High	Moderate	Translate an existing reliable English dataset.
2	Inadequate dataset translation	Poor model performance due to inaccurate data	Moderate	High	Manually review a subset of translated reviews for accuracy. If it is Inadequate, try a different approach.
3	Model underperformance	The model fails to achieve an acceptable level of accuracy or F1-score	Low	High	Continual model fine-tuning and consult with NLP experts.
4	Technical Challenges in Model Integration	Difficulties in integrating the selected NLP model into the system architecture may delay project timelines.	Moderate	Moderate	Engage with expert in NLP(supervisor) during the planning phase.
5	Runtime limitations	Interruptions in model training due to session timeouts or resource limits in Google Colab	High	High	Regularly save progress, plan for potential restarts, and split long-running tasks into smaller chunks
6	Loss of data due to system failure	Significant project delays.	Low	High	Regular data backups and use of robust data storage solutions
7	Resource Constraints	Insufficient resources, including time may impede the successful completion of project milestones.	Low	High	Develop contingency plans and prioritise tasks based on available resources.

Table 10 : Risk Assessment

Appendix E: Milestones

Milestone	Description	Completion Date
Problem Identification	Identified the need for an NLP system capable of detecting fake reviews in Telugu, highlighting the lack of existing tools for underrepresented languages.	11 Oct 2023
Literature Review	Comprehensive review of existing literature on NLP models for fake review detection, focusing on BERT and mBERT technologies, and exploring the challenges in adapting these models to new languages.	10 Nov 2023
Dataset Translation and Preparation	Translated the English OpSpam dataset into Telugu using Google Translate API and conducted manual reviews to assess translation quality.	17 Jan 2024
Model Adaptation and Training	Adapted and fine-tuned the BERT and mBERT models for the Telugu dataset, setting up experiments for transfer learning, zero-shot learning, and mixed-language training.	24 Feb 2024
Model Testing and Evaluation	Conducted extensive testing of the adapted models, evaluating their performance using accuracy, precision, recall, and F1-score metrics.	1 Mar 2024
Draft Report Compilation	Compiled the initial draft of the project report, documenting methodologies, results, and analyses.	22 Mar 2024
Final Report	Incorporated feedback from supervisors, refined the report, and prepared the final submission.	20 Apr 2024

Table 11: Milestones